

Brindlewythe Biosciences Ltd

AP02 hinge-binder triage: hERG and five-CYP panel, 40 analogues

Final Report

Study BW11_0275 (LIMS B-ACC-2011-04692)

Client: Avelnwick Pharmaceuticals Ltd

Study lead: Dr A. Veldershaan, Study Director

Report date: 27 September 2011

1 Summary

40 compounds from the Avelnwick aminopyrimidine regioisomer series 02 were tested for Avelnwick Pharmaceuticals Ltd in the hERG automated patch clamp, 5-isoform CYP panel. Testing started on 12 September 2011 and results were reported on 27 September 2011. All accepted runs met plate acceptance criteria; one run was rejected and repeated (see Section 6). No compound had a hERG IC₅₀ below 1 μ M. The set divided almost evenly around 10 μ M (20 below, 20 at or above 10 μ M or above the top concentration). Inhibition above 50% at 10 μ M was most frequent for CYP3A4; CYP1A2, CYP2C9 were rarely affected.

Table 1 Study summary

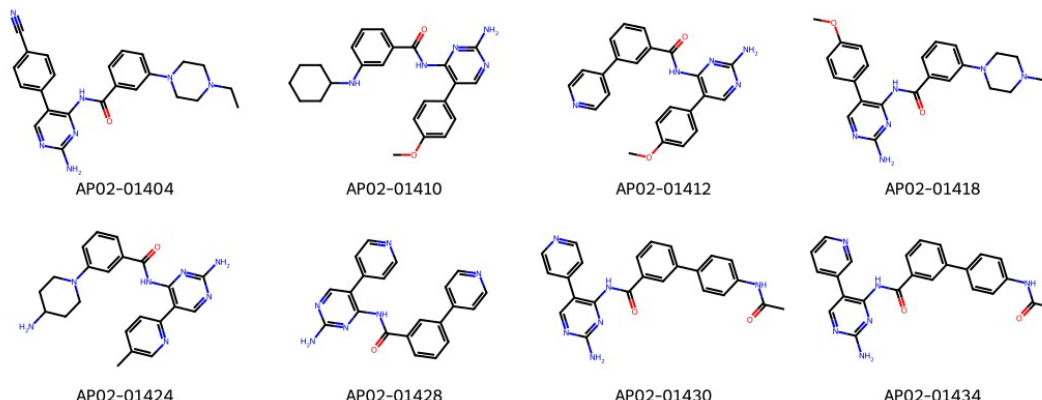
Metric	Value
Project	BW11_0275
Client	Avelnwick Pharmaceuticals Ltd
Compounds tested	40
Series / chemotype	Avelnwick aminopyrimidine regioisomer series 02 (Aminopyrimidine regioisomer)
Assays	hERG patch clamp, CYP inhibition panel (1A2/2C9/2C19/2D6/3A4) at 10 μ M
Turnaround (calendar days)	15
Price	GBP 11,800
hERG IC ₅₀ <1 μ M	0
hERG IC ₅₀ 1-10 μ M	20
hERG IC ₅₀ $\geq$ 10 μ M	20
hERG median IC ₅₀ (μ M)	10
E-4031 control IC ₅₀ (nM)	16
CYP1A2 >50% inhibition @10 μ M	2
CYP2C9 >50% inhibition @10 μ M	2
CYP2C19 >50% inhibition @10 μ M	9
CYP2D6 >50% inhibition @10 μ M	7
CYP3A4 >50% inhibition @10 μ M	11
Compounds with any CYP >50%	16
Ketoconazole (3A4) IC ₅₀ (μ M)	0.029

2 Objective

To rank the screening set for hERG automated patch clamp, 5-isoform CYP panel liability so that Avelon Pharmaceuticals Ltd could choose analogues for progression.

3 Test items

40 compounds were received as 10 mM DMSO solutions in one test lot on 12 September 2011 and logged against B-ACC-2011-04692. Identity and concentration were taken as supplied by the client. Stocks were stored at -20 °C and thawed once for testing.



4 Methods

hERG automated patch clamp

HEK293 cells stably expressing hERG (Kv11.1) were recorded in whole-cell configuration on a 16-channel automated patch clamp platform at room temperature. Cells are held at -80 mV, depolarised to +40 mV for 2 s and repolarised to -50 mV for 2 s to elicit the tail current, at 0.1 Hz. Each compound was applied cumulatively at 0.12, 0.37, 1.1, 3.3, 10 and 30 μ M (0.3% DMSO final). Each concentration is held until the tail-current block reaches a steady state (change below 5% over the last three sweeps), with a minimum of 3 and a maximum of 8 minutes; compounds still drifting at 8 minutes are flagged as slow-onset.

E-4031 is run as a separate 6-point concentration response (1-300 nM) on every run.

- Plate acceptance: seal resistance ≥ 500 M Ω before break-in and ≥ 200 M Ω throughout
- Tail current ≥ 200 pA at baseline; run-down $\leq 10\%$ over the vehicle period
- E-4031 IC₅₀ within 5-50 nM; runs outside this range are rejected and repeated
- IC₅₀ fitted per cell (four-parameter logistic); the compound IC₅₀ is the geometric mean of finite cell IC₅₀s when these are the majority of accepted cells, otherwise reported as >30 μ M
- Where precipitation is seen at 30 μ M, that concentration is excluded and the highest evaluable concentration is reported

5-isoform CYP panel

Pooled human liver microsomes (0.1 mg/mL protein) were incubated with each compound at 10 μ M and an isoform-selective probe substrate at approximately K_m , in duplicate, for 10 minutes at 37 °C after

NADPH addition. Metabolite formation is measured by LC-MS/MS against vehicle (0.1% DMSO) incubations; the lower limit of quantitation of each metabolite is at least 20-fold below the vehicle signal.

Probe substrates and controls

Isoform	Probe substrate	Metabolite	Positive control
CYP1A2	Phenacetin	Acetaminophen	Furafylline
CYP2C9	Diclofenac	4'-Hydroxydiclofenac	Sulfaphenazole
CYP2C19	S-Mephenytoin	4'-Hydroxymephenytoin	Ticlopidine
CYP2D6	Dextromethorphan	Dextrorphan	Quinidine
CYP3A4	Midazolam	1'-Hydroxymidazolam	Ketoconazole

Positive controls are run as 7-point concentration responses on every run. Furafylline and ticlopidine (mechanism-based inhibitors) are preincubated with microsomes and NADPH for 15 minutes before probe addition. A run is accepted when each control IC₅₀ falls within 0.5-2 times the laboratory historical mean. Results are reported as mean percent inhibition relative to vehicle; values above 50% at 10 μ M are flagged, and small negative values indicate no inhibition.

5 Results

hERG results

All 40 compounds were tested. No compound had a hERG IC₅₀ below 1 μ M. The set divided almost evenly around 10 μ M (20 below, 20 at or above 10 μ M or above the top concentration). The median IC₅₀ was 10 μ M (0 below 1 μ M, 20 between 1 and 10 μ M, 20 at or above 10 μ M, of which 3 above the highest evaluable concentration). E-4031 gave an IC₅₀ of 16 nM in the accepted runs.

Compounds with QC notes: AP02-01434: First run rejected (E-4031 outside range); repeated; AP02-01436: First run rejected (E-4031 outside range); repeated; AP02-01572: First run rejected (E-4031 outside range); repeated. Values for these compounds should be read with the note.

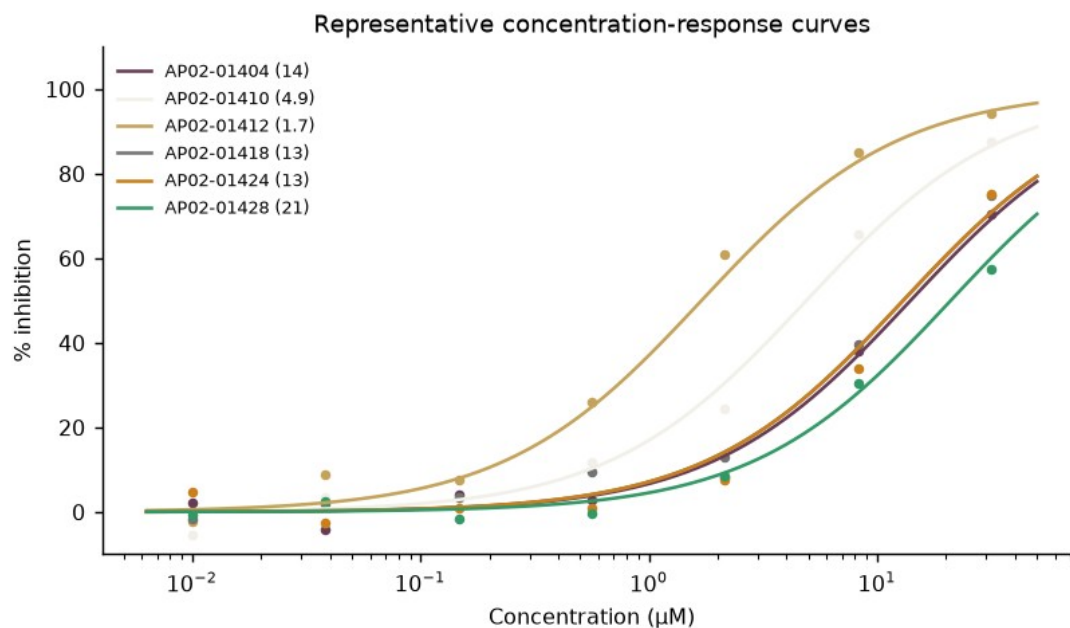
Table 2 hERG IC₅₀ by compound (μ M)

Compound	IC ₅₀ cell 1 (μ M)	IC ₅₀ cell 2 (μ M)	IC ₅₀ cell 3 (μ M)	IC ₅₀ geomean (μ M)	% inhibition @10 μ M	n cells (finite)	QC note
AP02-01404	14	13	12	13	51	3 (3)	
AP02-01410	4.9	5.3	3.0	4.3	68	3 (3)	
AP02-01412	1.7	1.2		1.4	86	2 (2)	
AP02-01418	13	10	12	12	52	3 (3)	
AP02-01424	13	21		17	37	2 (2)	
AP02-01428	21	>30		>30	28	2 (1)	
AP02-01430	7.6	11		9.2	61	2 (2)	
AP02-01434	16	23		19	36	2 (2)	First run rejected (E-4031 outside range); repeated
AP02-01436	16	11		13	46	2 (2)	First run rejected (E-4031 outside range); repeated
AP02-01442	13	14		13	46	2 (2)	
AP02-01454	5.8	4.4		5.0	69	2 (2)	
AP02-01458	4.0	4.9		4.5	69	2 (2)	
AP02-01464	13	8.4		11	49	2 (2)	

AP02-01466	6.4	6.4	8.1	6.9	54	3 (3)	
AP02-01472	22	24		23	29	2 (2)	
AP02-01482	4.6	4.3		4.5	66	2 (2)	
AP02-01484	10	17	13	13	39	3 (3)	
AP02-01490	16	27		21	29	2 (2)	
AP02-01494	3.3	2.6		2.9	80	2 (2)	
AP02-01496	16	24		20	30	2 (2)	
AP02-01500	5.4	6.5		5.9	68	2 (2)	
AP02-01502	6.0	7.3		6.6	52	2 (2)	
AP02-01506	29	>30		>30	21	2 (1)	
AP02-01508	8.0	11		9.2	52	2 (2)	
AP02-01514	19	20		19	30	2 (2)	
AP02-01520	25	23		24	25	2 (2)	
AP02-01524	22	>30		>30	24	2 (1)	
AP02-01532	3.7	6.1		4.7	74	2 (2)	
AP02-01536	4.8	3.2	3.9	3.9	75	3 (3)	
AP02-01538	16	11		14	46	2 (2)	
AP02-01542	8.1	10		9.0	57	2 (2)	
AP02-01544	17	14		15	43	2 (2)	
AP02-01548	2.5	2.5	2.7	2.6	78	3 (3)	
AP02-01556	6.1	6.4		6.2	58	2 (2)	
AP02-01560	9.5	6.8		8.0	58	2 (2)	
AP02-01568	10	7.5		8.8	50	2 (2)	
AP02-01572	15	6.4		9.6	49	2 (2)	First run rejected (E-4031 outside range); repeated
AP02-01574	20	18		19	31	2 (2)	
AP02-01584	8.2	11		9.4	49	2 (2)	
AP02-01590	12	10		11	47	2 (2)	

Top concentration 30 μ M; 6-point, 3-fold dilution; positive control E-4031 IC₅₀ 16 nM (acceptance 5-50 nM)

Compound IC₅₀ = geometric mean of finite cell IC₅₀s when they are the majority of accepted cells; otherwise reported as >30 μ M (or >10 μ M where solubility-limited)



CYP inhibition results

Inhibition above 50% at 10 µM was most frequent for CYP3A4; CYP1A2, CYP2C9 were rarely affected. In total 16 compounds inhibited at least one isoform by more than 50% at 10 µM (CYP3A4: 11; CYP2D6: 7; CYP2C19: 9; CYP2C9: 2; CYP1A2: 2). All control IC₅₀s were within 0.5-2 times the historical mean (ketoconazole 0.029 µM against a historical mean of 0.030 µM).

Table 3 Percent inhibition at 10 µM

Compound	CYP1A2 (%)	CYP2C9 (%)	CYP2C19 (%)	CYP2D6 (%)	CYP3A4 (%)
AP02-01404	-2	4	2	-1	10
AP02-01410	4	34	91	95	85
AP02-01412	54	82	95	46	97
AP02-01418	-1	5	13	7	13
AP02-01424	3	10	7	1	-7
AP02-01428	68	4	50	-0	36
AP02-01430	11	-0	40	17	54
AP02-01434	12	-1	11	17	85
AP02-01436	4	4	3	10	2
AP02-01442	-3	28	67	-9	33
AP02-01454	10	13	39	13	25
AP02-01458	20	48	69	53	22
AP02-01464	1	8	-4	1	4
AP02-01466	-5	9	16	41	15
AP02-01472	-4	-5	7	17	45
AP02-01482	-2	5	43	87	15
AP02-01484	-1	9	15	4	8
AP02-01490	9	3	-0	-4	-3
AP02-01494	37	11	69	10	82
AP02-01496	-6	1	9	6	4
AP02-01500	7	1	5	19	-3
AP02-01502	-10	-3	10	-3	6
AP02-01506	4	3	-3	-9	-5
AP02-01508	-1	40	5	99	61
AP02-01514	5	3	32	6	6
AP02-01520	-1	3	21	4	46
AP02-01524	18	7	1	22	11
AP02-01532	-2	8	9	71	27
AP02-01536	12	3	71	71	70

AP02-01538	1	-1	6	45	1
AP02-01542	-1	2	1	3	4
AP02-01544	33	3	63	53	60
AP02-01548	1	46	40	7	66
AP02-01556	17	70	92	48	73
AP02-01560	8	4	7	4	-4
AP02-01568	8	8	2	40	27
AP02-01572	2	-4	7	29	20
AP02-01574	39	3	41	-1	80
AP02-01584	7	-3	-8	12	35
AP02-01590	7	6	33	9	36

Single concentration 10 μ M, duplicate incubations, mean reported; negative values = no inhibition

Controls (IC₅₀, μ M): furafylline (CYP1A2) 2.3; sulfaphenazole (CYP2C9) 0.33; ticlopidine (CYP2C19) 1.0; quinidine (CYP2D6) 0.060; ketoconazole (CYP3A4) 0.029

Table 4 Positive control IC₅₀s

Control	Isoform	IC ₅₀ (μ M)	Historical mean (μ M)
Furafylline	CYP1A2	2.3	1.8
Sulfaphenazole	CYP2C9	0.33	0.35
Ticlopidine	CYP2C19	1.0	1.2
Quinidine	CYP2D6	0.060	0.060
Ketoconazole	CYP3A4	0.029	0.030

6 Deviations

The first hERG run for one compound was rejected because E-4031 fell outside its acceptance range; the compound was retested in a later run and only accepted data are reported.

7 Discussion

No compound had a hERG IC₅₀ below 1 μ M. The set divided almost evenly around 10 μ M (20 below, 20 at or above 10 μ M or above the top concentration). Inhibition above 50% at 10 μ M was most frequent for CYP3A4; CYP1A2, CYP2C9 were rarely affected. These are screening data; mechanism statements are outside the scope of this study.

8 Conclusion

Recommended next steps: prioritise compounds with hERG IC₅₀ at or above 10 μ M; for any compound below 10 μ M, compare the IC₅₀ with projected free plasma exposure before progression; repeat the flagged hERG measurements if a selection decision depends on them; run concentration responses for compounds flagged on CYP3A4 and any other isoform above 50%.

9 Records

Raw data, run sheets and instrument files are retained for 5 years under B-ACC-2011-04692.

Name	Role	Signature	Date
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Anika Veldershain	Study Director	<u>A Veldershain</u>	27 September 2011
Ines Alvarado	Assay Quality Coordinator	<u>I Alvarado</u>	26 September 2011